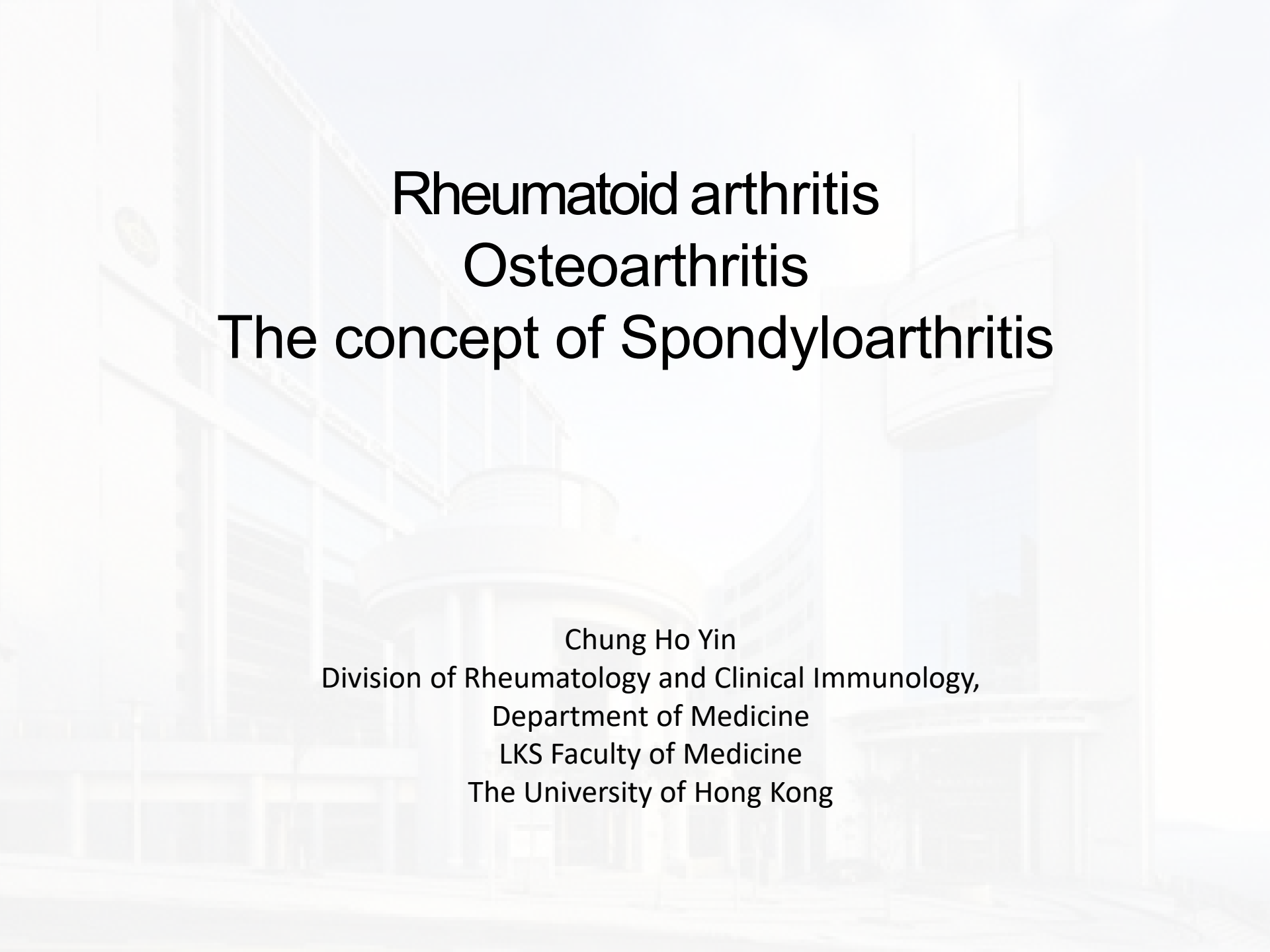




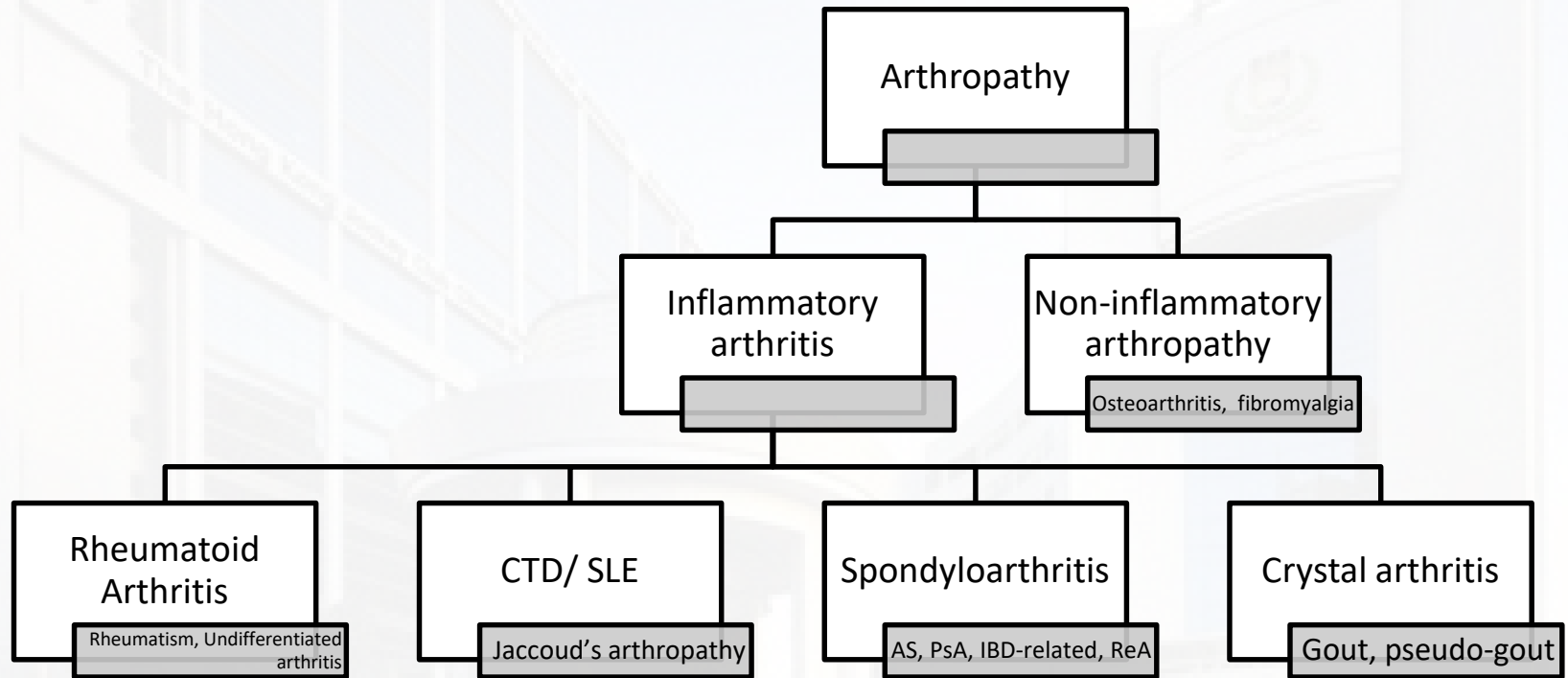
Rheumatoid arthritis Osteoarthritis The concept of Spondyloarthritis

Chung Ho Yin
Division of Rheumatology and Clinical Immunology,
Department of Medicine
LKS Faculty of Medicine
The University of Hong Kong

Outline of lecture

- Approach to joint pain
- Rheumatoid Arthritis
- Osteoarthritis
- The concept of Spondyloarthritis
- Disease management

Polyarthritis in rheumatology



Clinical features of different arthritis --- history

Rheumatoid Arthritis

- Younger age
- Insidious onset
- Polyarthritis
- Small hand joints
- Sparing DIP joints
- Early morning stiffness ≥ 30 mins

Spondyloarthritis

- Younger age
- Insidious onset
- Mono or polyarthritis
- Any joint could be involved
- Psoriasis, IBD, STD/ dysentery, uveitis
- Back pain
- Family history

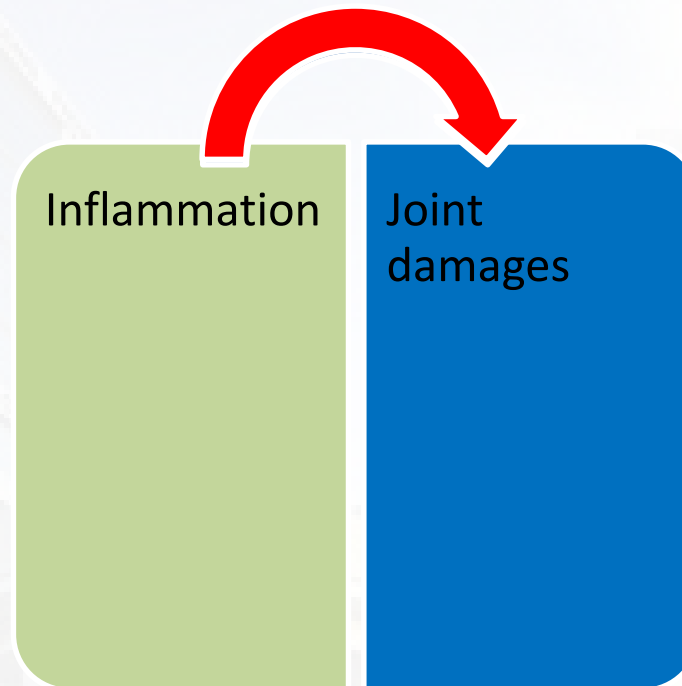
Osteoarthritis

- Insidious onset
- DIP joints or weight bearing joints
- Older age

Gout

- Acute onset
- First metatarsophalangeal joint involvement
- Usually self limiting
- Sometimes fever

Concept of arthritis



Arthritis --- physical examination



Arthritis --- physical examination

DIP joints



MCP joints



MTP joint



Knee



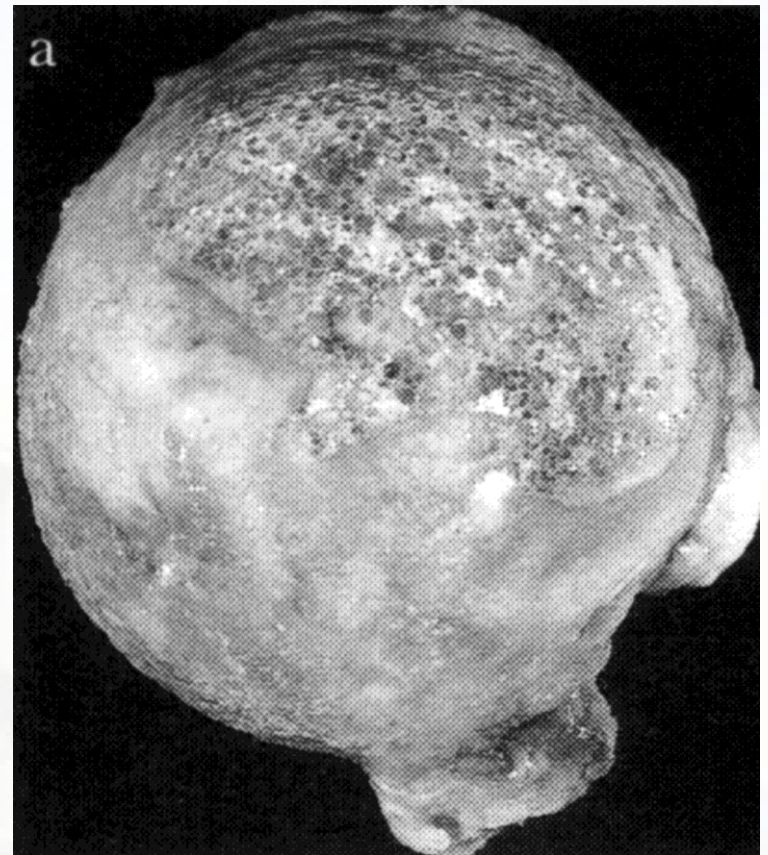
Osteoarthritis (OA)

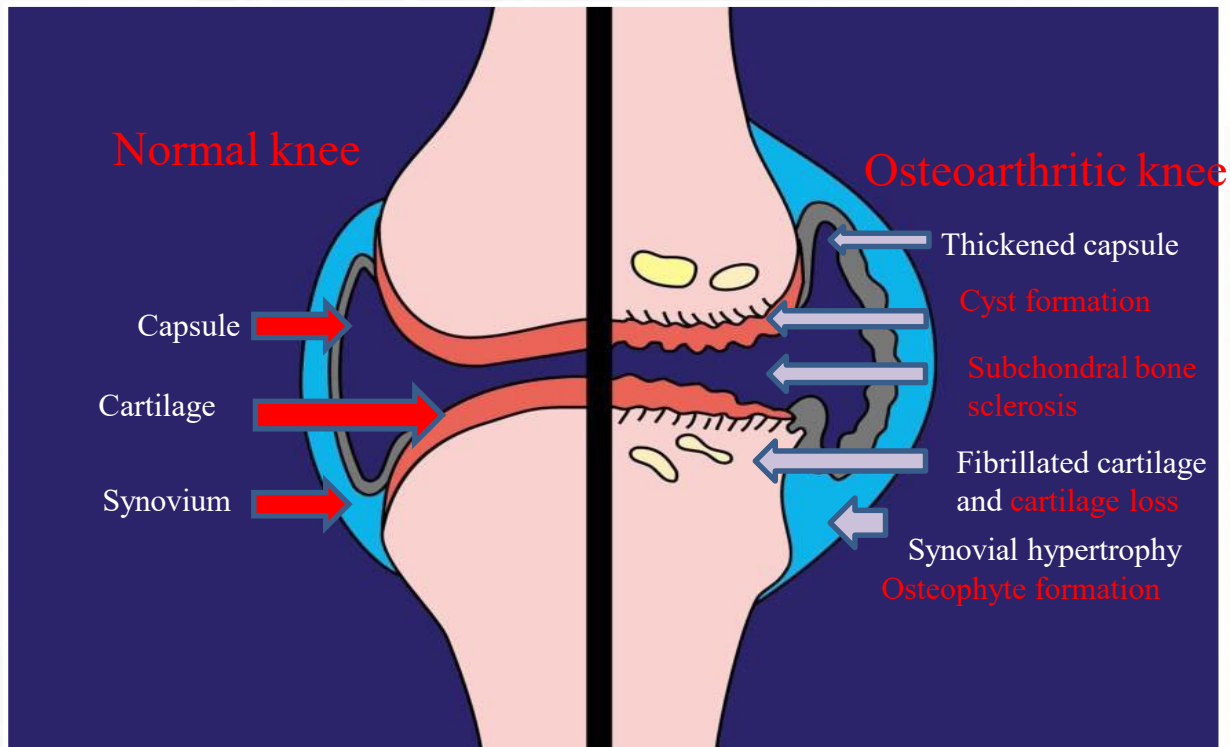
- Commonest cause of non-inflammatory arthropathy
- Prevalence increases steeply with age
 - *Affects 75% of women $\geq$ age 65*
- Sites:
 - *Weight bearing joints: knees, hips, spine*
 - *CMC joint*
 - *DIP joint*
 - *Trapeziometacarpal joints*
- Three or more joints involved: generalized OA
- Physical exam: crepitus
- Primary OA vs secondary OA



Cartilage surface of a normal femoral head

Cartilage surface of an osteoarthritic femoral head





Osteoarthritis

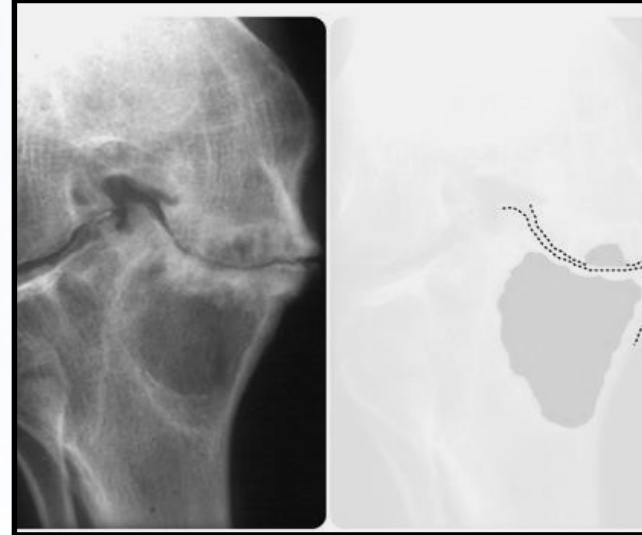
Imaging features

- preserved bone density
- joint space narrowing (cartilage destruction)
- sclerosis of subchondral bone
- subchondral cysts
- marginal osteophytes
- joint deformity

Sub-chondral sclerosis



Sub-chondral cyst



Narrowing of joint space



**Radiological features of
osteoarthritis**

Rheumatoid Arthritis

- The Commonest cause of autoimmune inflammatory polyarthritis
- It is found in all races
- Disease prevalence:
 - 1-2% in Caucasians
 - 0.3-0.4% in Chinese
 - Rare in blacks
- Peak age of onset: 35-55 years
- Female to male ratio = 3:1



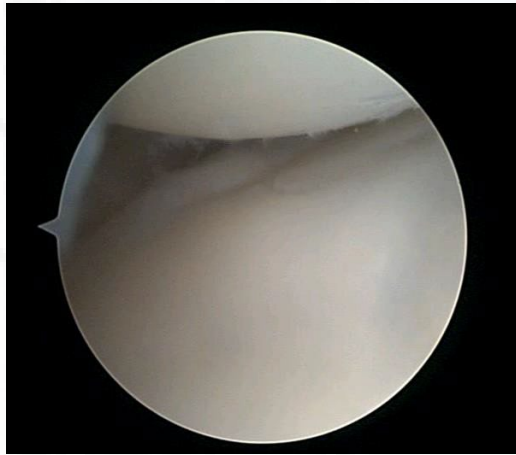
Rheumatoid
Arthritis

- One of the commonest autoimmune inflammatory arthritis
- Prevalence varies:
 - 1% to 2% in Caucasians
 - 0.3% to 0.4% in Chinese
- Peak age of onset: 35-55 years old
- Female to male ratio: 3:1
- Insidious onset of symptoms



Rheumatoid Arthritis

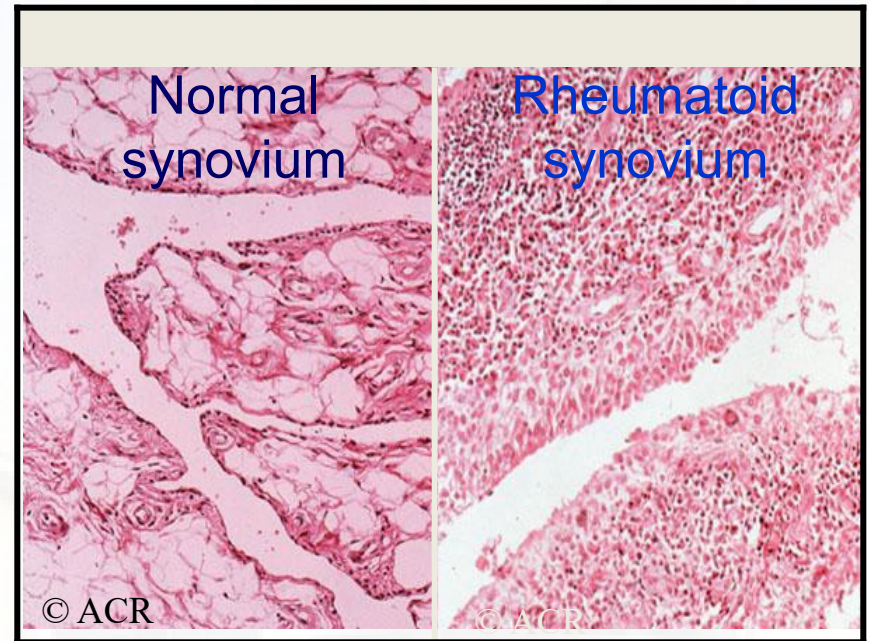
Synovial proliferation and inflammation in rheumatoid arthritis



Arthroscopic appearance of a **normal** joint



Arthroscopic appearance of a **rheumatoid** joint



Histological appearances of **normal** and **rheumatoid** synovium

Radiological features of RA

- **Sites**

- appendicular

- hands and feet - MCP and PIP, earliest changes in 2nd & 3rd MCP and 3rd PIP
 - wrists, elbows, knees, ankles, shoulders , hips

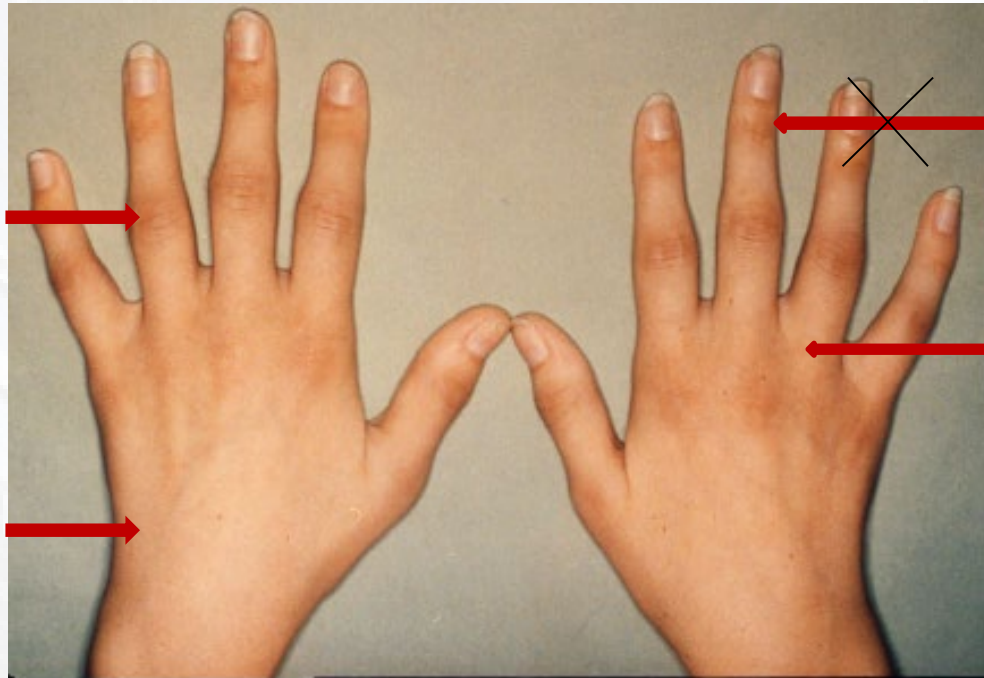
- axial skeleton

- apophyseal and atlantoaxial joints of cervical spine

Pattern of joint involvement in Rheumatoid Arthritis

Proximal
interphalangeal (PIP)
joint

Wrist joint



Distal interphalangeal
(DIP) joint

Metacarpophalangeal
(MCP) joint

Features of Rheumatoid Arthritis

Disease activities

- Pain
- Swelling
- Stiffness (morning)
- Functional limitation

Clinical: DAS score
Biomarkers: CRP/ESR
Ultrasound assessment
MRI assessment

Disease damages

- Pain
- Deformities:
subluxation
(including C1/2),
tendon rupture,
dislocation...
- Secondary OA
- Functional limitation

X-rays
CT

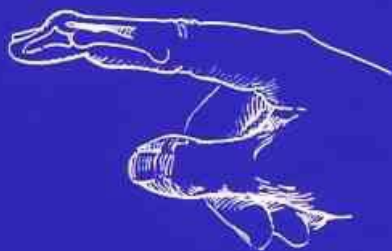
Extra-articular features



Swan neck deformity

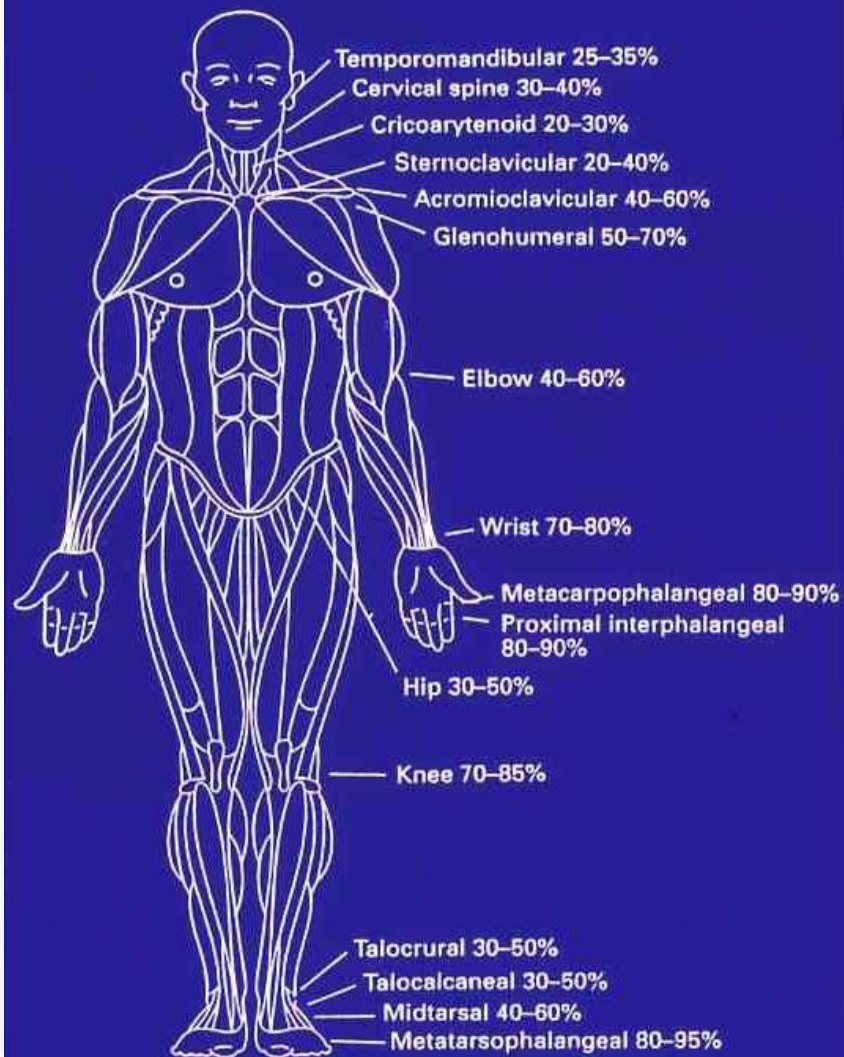


Buttonnaire deformity

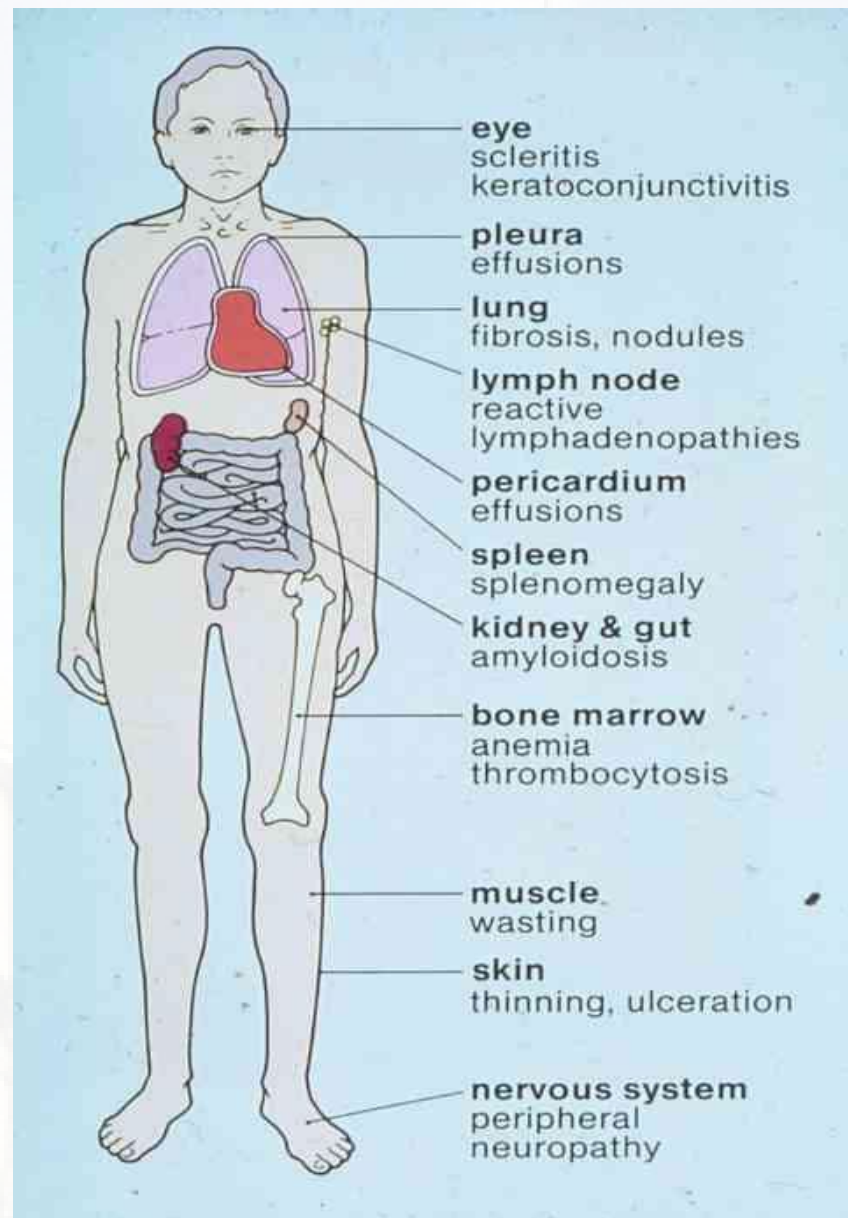


Mallet finger





Joint involvement in long-standing rheumatoid arthritis.



The 2010 ACR/EULAR classification criteria for rheumatoid arthritis

Target population (Who should be tested?): Patients who

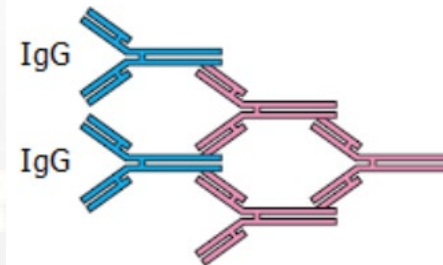
- 1) have at least 1 joint with definite clinical synovitis (swelling)*
- 2) with the synovitis not better explained by another disease†

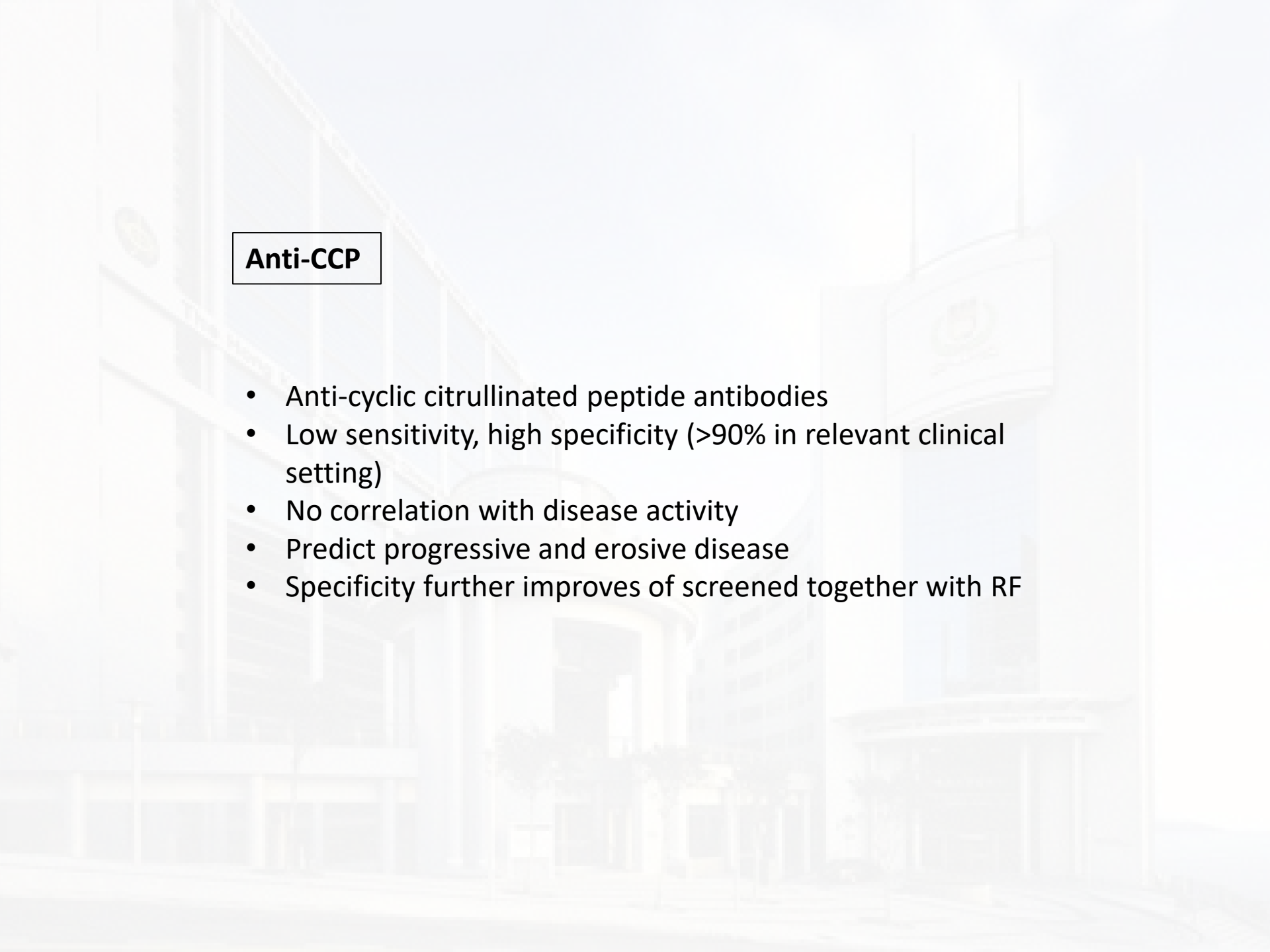
Classification criteria for RA (score-based algorithm: add score of categories A–D;
a score of $\geq 6/10$ is needed for classification of a patient as having definite RA)‡

A. Joint involvement§	
1 large joint¶	0
2–10 large joints	1
1–3 small joints (with or without involvement of large joints)#	2
4–10 small joints (with or without involvement of large joints)	3
>10 joints (at least 1 small joint)**	5
B. Serology (at least 1 test result is needed for classification)††	
Negative RF <i>and</i> negative ACPA	0
Low-positive RF <i>or</i> low-positive ACPA	2
High-positive RF <i>or</i> high-positive ACPA	3
C. Acute-phase reactants (at least 1 test result is needed for classification)‡‡	
Normal CRP <i>and</i> normal ESR	0
Abnormal CRP <i>or</i> abnormal ESR	1
D. Duration of symptoms§§	
<6 weeks	0
≥ 6 weeks	1

Rheumatoid factor

- Auto-antibody against IgG
- Not sensitivity
- No correlation with disease activity
- Causes:
 - 1. Rheumatoid Arthritis
 - 2. Sjogren's Syndrome
 - 3. Chronic infection
 - 4. elderly and healthy individuals
- High levels in rheumatoid arthritis => poor prognosis



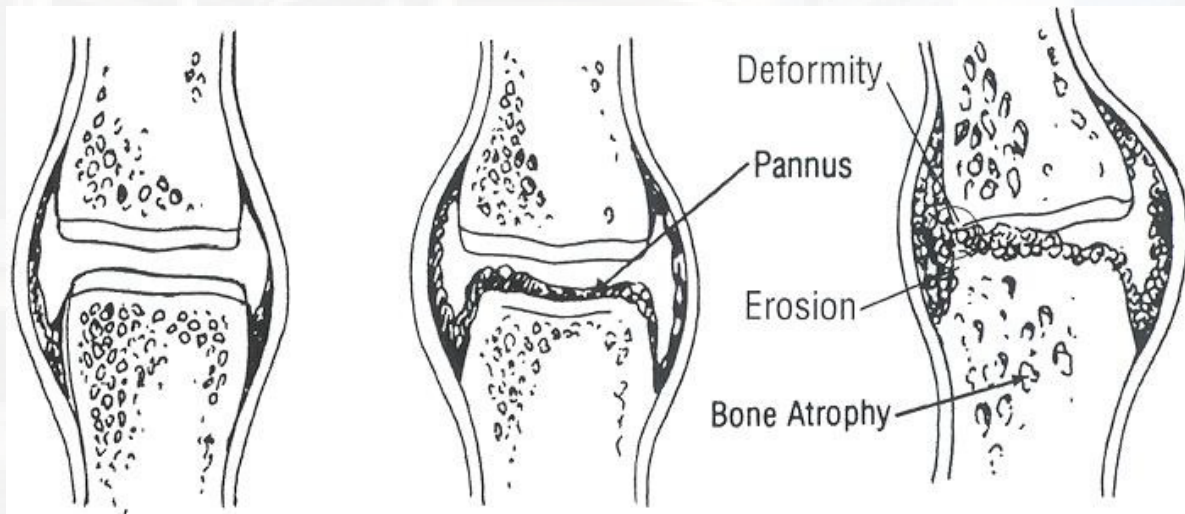


Anti-CCP

- Anti-cyclic citrullinated peptide antibodies
- Low sensitivity, high specificity (>90% in relevant clinical setting)
- No correlation with disease activity
- Predict progressive and erosive disease
- Specificity further improves if screened together with RF

Radiological features of RA

- Soft tissue swelling and widened joint space
 - synovial inflammation and effusion
- Juxta-articular osteoporosis
 - hyperaemia and disuse
- Joint space narrowing
 - destruction of cartilage by pannus
- Periarticular erosions
 - pannus destruction of unprotected bone at insertion of joint capsule



Rheumatoid arthritis



Rheumatoid arthritis



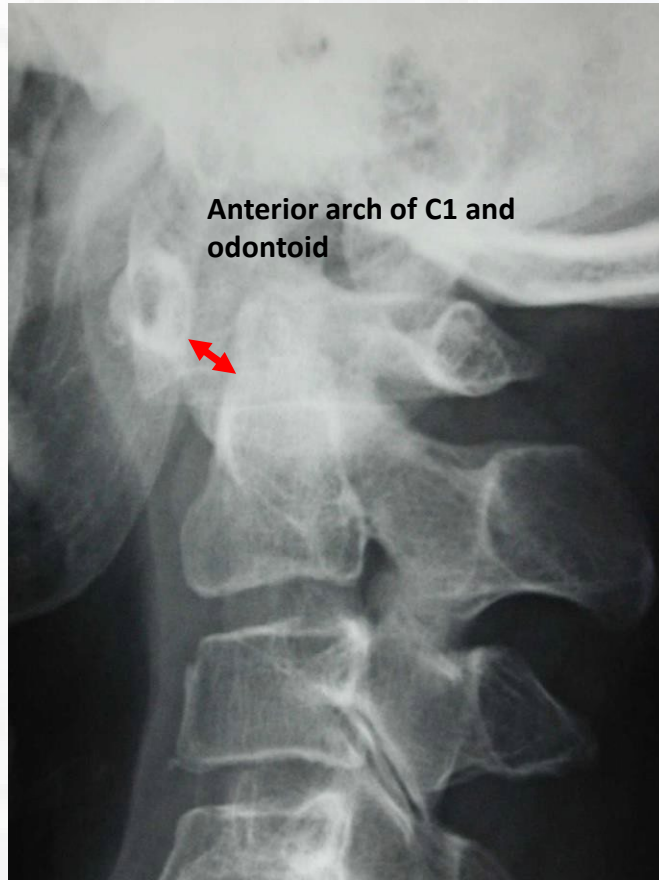
Rheumatoid arthritis



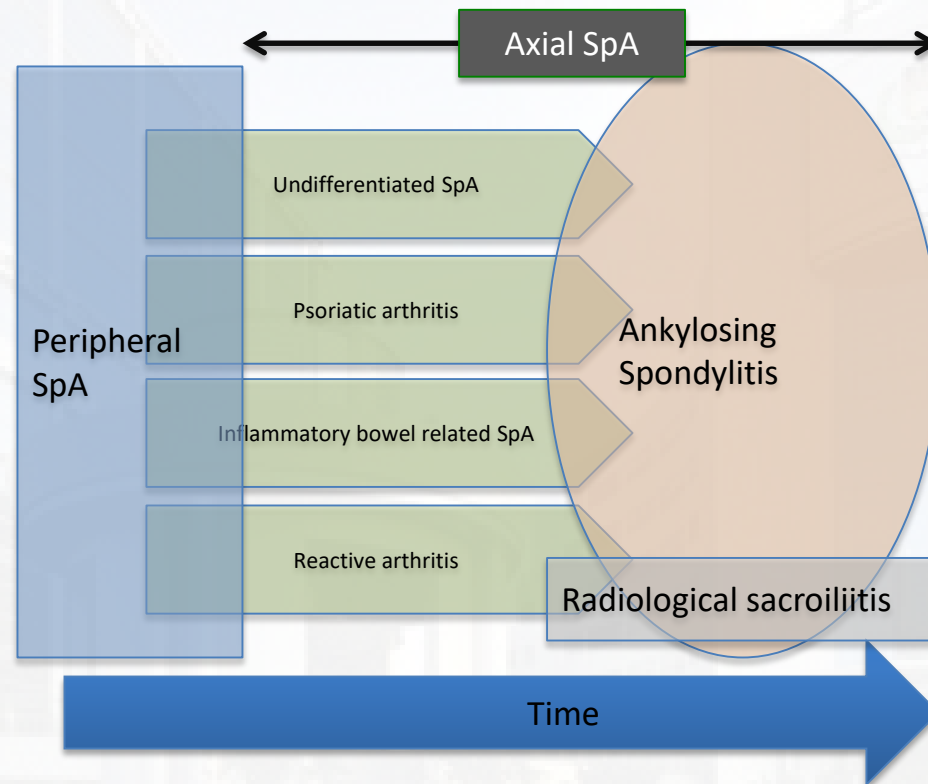
Rheumatoid arthritis



Rheumatoid arthritis



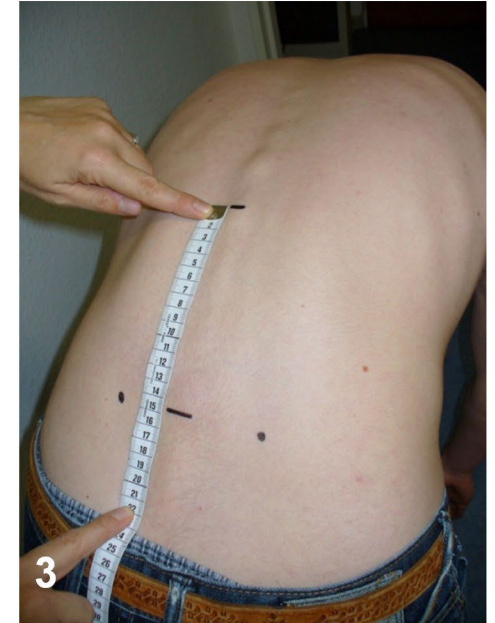
Concept of Spondyloarthritis (SpA)



Clinical features of SpA

- Spondylitis
- Peripheral arthritis
- Enthesitis (Achilles tendinitis, plantar fasciitis)
- Anterior uveitis
- Aortitis
- Associated with HLA-B27
- Other associated features: psoriasis, inflammatory bowel disease, dysentery, sexually transmitted disease

Spinal Mobility – Modified Schober



- Patient standing erect
- Mark an imaginary line connecting both posterior superior iliac spines (close to the dimples of Venus) (1)
- A next mark is placed 10 cm above (2)
- The patient bends forward maximally, measure the difference between the two marks (3)
- Report the increase (in cm to the nearest 0.1 cm)
- The best of two tries is recorded.

Spinal Mobility – Lateral Spinal Flexion



- Heels and back rest against the wall. No flexion in the knees, no bending forward
- Place a mark on the thigh (1), bend sideways without bending knees or lifting heels (2), and without moving the shoulders or hips, place a second mark and record the difference (3)
- The best of two tries is recorded for left and right separately.
- Finally, the mean of left and right is calculated (in cm to the nearest 0.1 cm).

Spinal Mobility - Occiput to Wall (black arrow) and Tragus to Wall (white arrow)



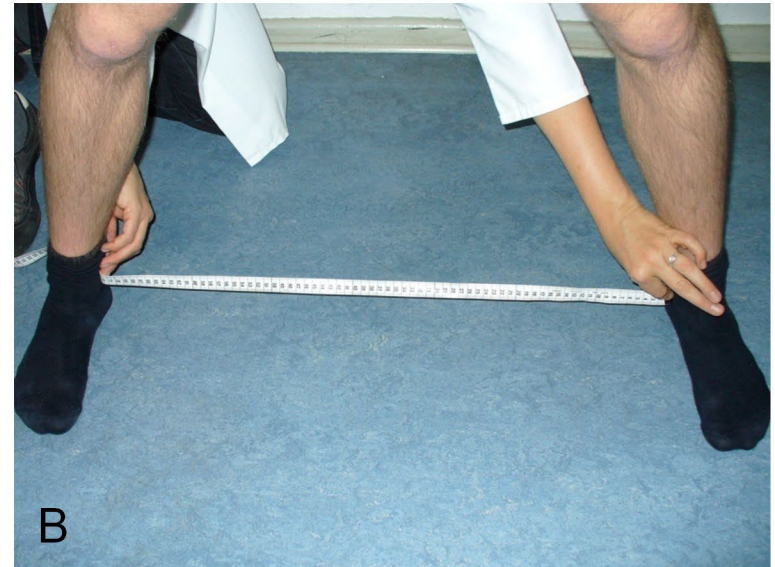
- Heels and back rest against the wall
- Chin at usual carrying level
- Maximal effort to move the head (occiput) against the wall
- Report the best of two tries (in cm) for the occiput to wall distance and the mean of left and right for the tragus to wall distance

Spinal Mobility – Cervical Rotation



- The patient sits straight on a chair, chin at usual carrying level, hands on the knees.
- The assessor places a goniometer at the top of the head in line with the nose (1.)
- The assessor asks to rotate the neck maximally to the left, follows with the goniometer, and records the angle between the sagittal plane and the new plane after rotation (2.)
- A second reading is taken and the best of the two is recorded for the left side
- The procedure is repeated for the right side.
- The mean of left and right is recorded in degrees (0-90 Degree).

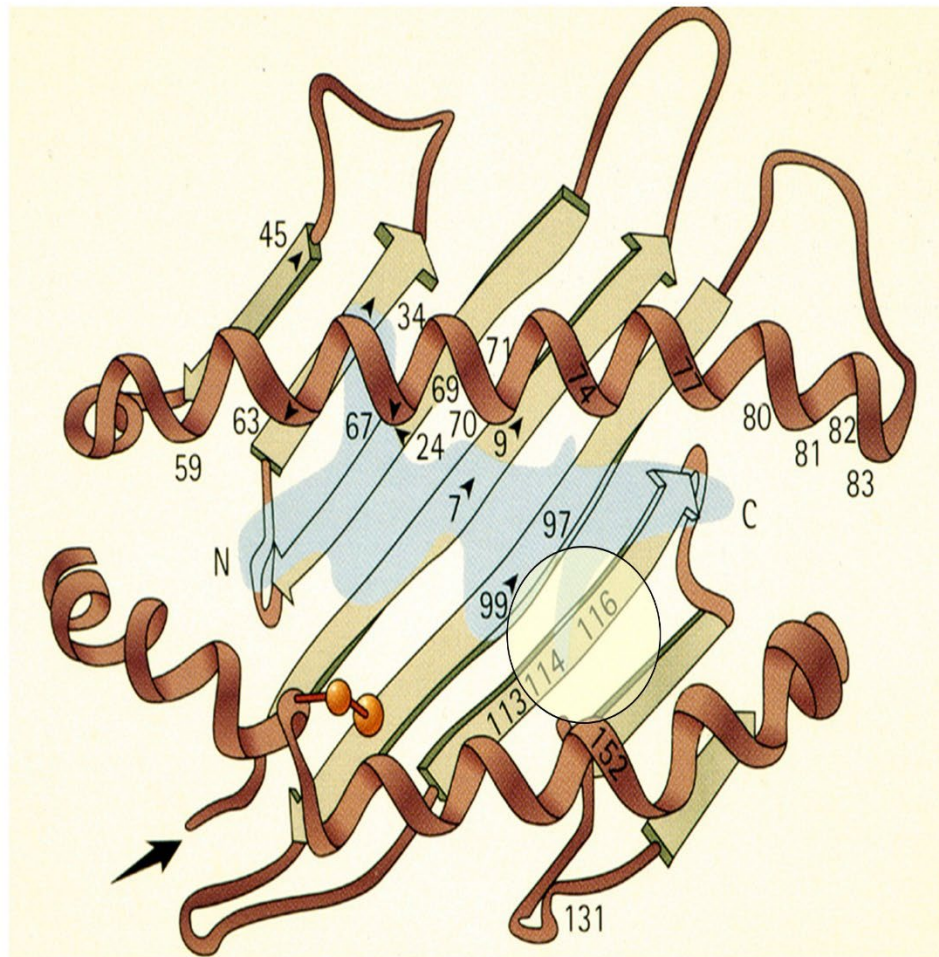
Intermalleolar Distance



- The patient is lying down (A) with the legs separated as far as possible with knees straight and toes pointing upwards (preferred method).
- Alternatively, the patient stands (B) and separates the legs as far as possible.
- The distance between the medial malleoli is measured.

Peptide-Binding Groove of HLA-B27

Disease predisposing B27 subtype *04 differs just in positions 114 and 116 from non-predisposing subtype *06



Khan MA: Spondyloarthropathies. In: Hunder G (Ed.). ATLAS OF RHEUMATOLOGY. 4rd Edition. Philadelphia, PA: Current Medicine 2005, pp. 151-180.(with permission)

HLA-B27: diagnosis and prognosis

- First reported in 1973
- Estimated 8-9% of Caucasian population carry HLA-B27 gene₁
- Positive in 90-95% of AS patients₂
- Positive in 46-75% of axial SpA patients_{3,4}
- HLA-B27 is not sensitive or specific in SpA diagnosis.
- It has a prognostic value.

1. David CS. The mystery of HLA-B27 and disease. *Immunogenetics*. 1997; 46:73-77.

2. Sheehan NJ. The ramifications of HLA-B27. *J R Soc Med* 2004; 97:10-4.

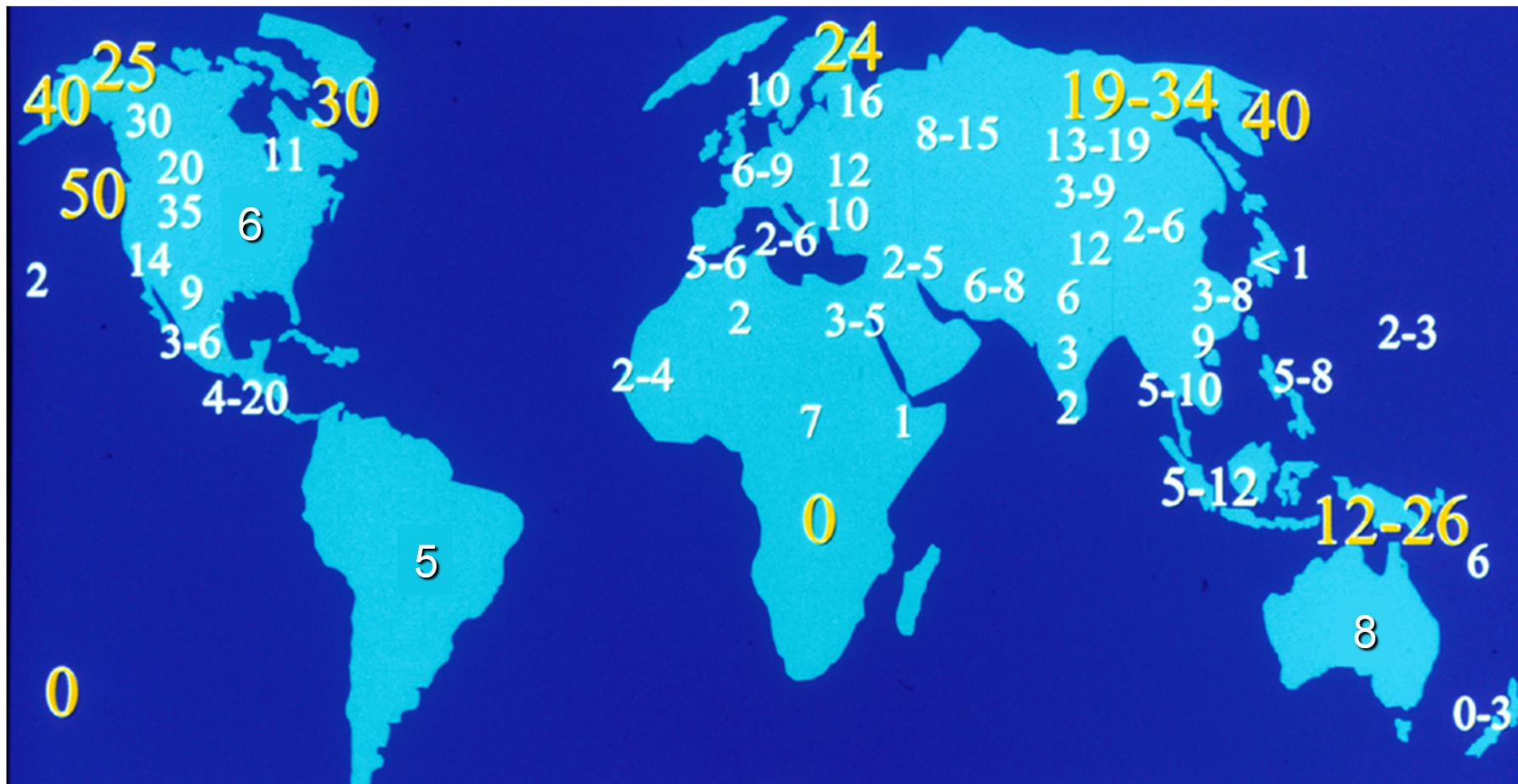
3. Rudwaleit M, Haibel H, Baraliakos et al. The early disease stage in axial spondylarthritis: result from the German Spondyloarthritis Inception Cohort. *Arthritis Rheum* 2009; 60:717-27.

4. Heuft-Dorenbosch L, Landewé R, Weijers R, *et al.*. Performance of various criteria sets in patients with inflammatory back pain of short duration; the Maastricht early spondyloarthritis clinic. *Ann Rheum Dis* 2007;66:92-8.



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Percentage Prevalence of HLA-B27 in Various Populations of the World



Khan MA Curr Opin Rheumatol 1995;7:263-9

Khan MA J Clin Rheumatol 2008;14:50-2

Khan MA. In Mehra N (Ed). The HLA Complex in Biology and Medicine. New Dehli, India 2010; 422-46.

Reveille J et al. Arthritis Rheum 2012;64:1407-11

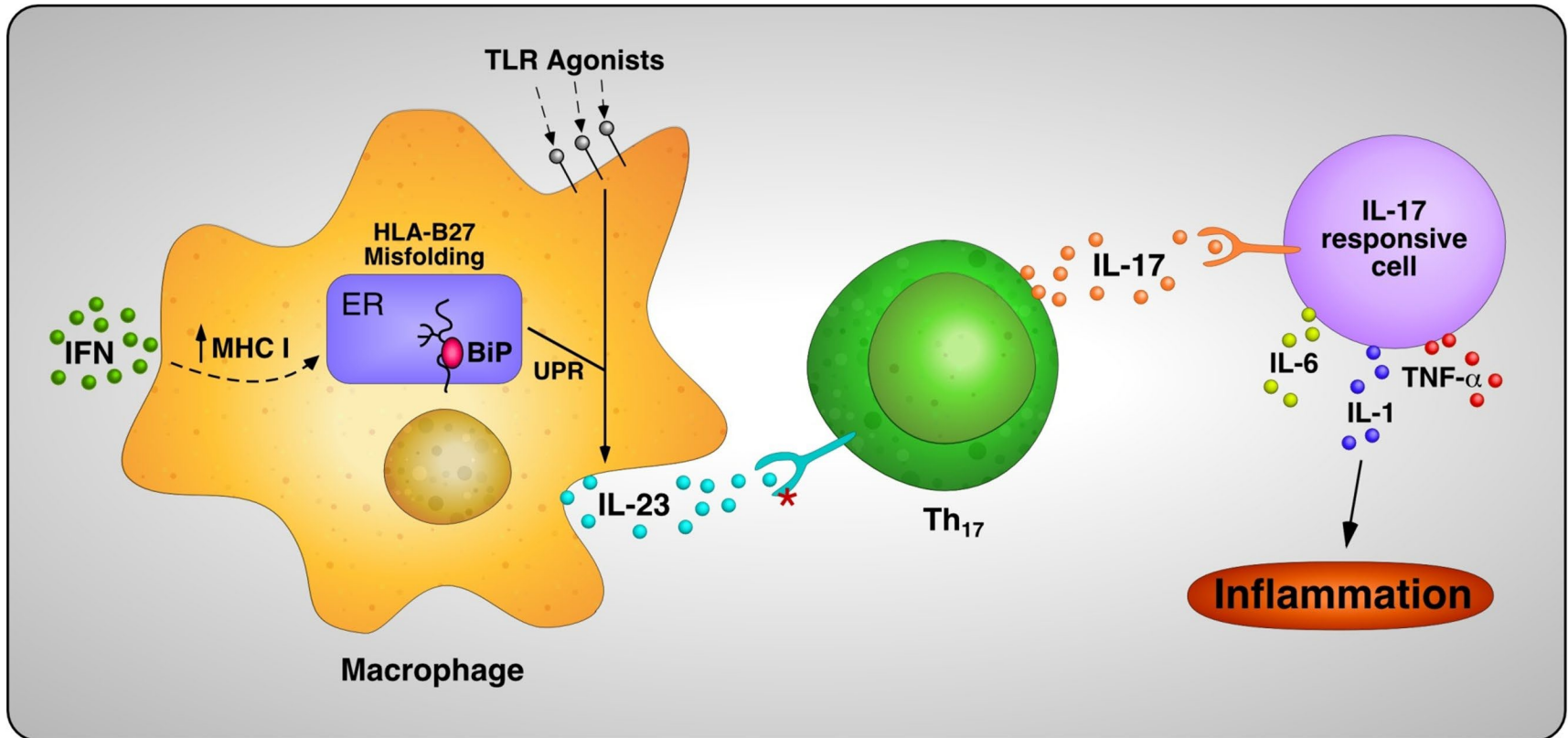


HLA-B27

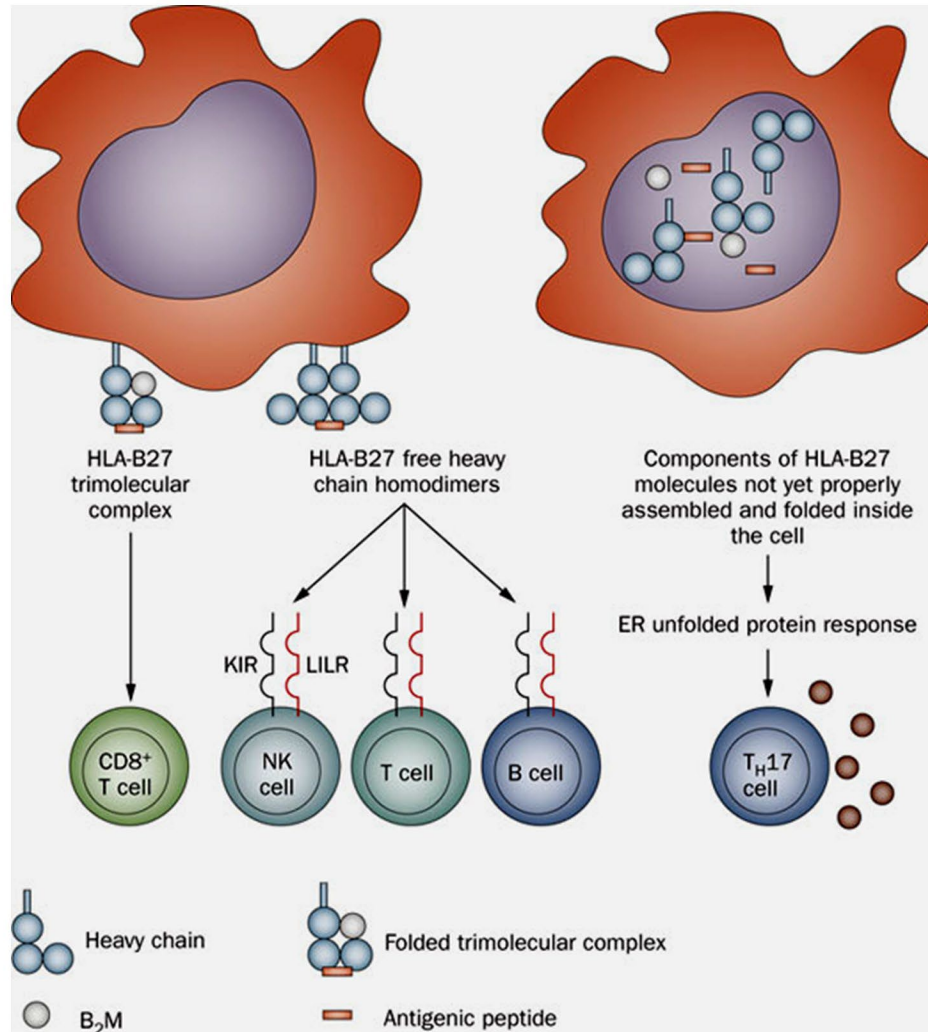
- 3 theories:
 1. B27 presenting an “arthritogenic” or “enthesitogenic” peptide to CD8+ T cells
 2. B27 heavy chain tends to misfold during synthesis in the endoplasmic reticulum (ER) and provoke an ER stress response. This influences cytokine secretion (e.g. IL-23)
 3. B27 expressed on the cell surface as dimers/multimers that engage receptors on T cells, NK cells and antigen-presenting cells (APCs) and affect their functioning.

HLA-B27 Misfolding & the IL-23/IL-17 Axis

Possible Consequences of HLA-B27 Misfolding



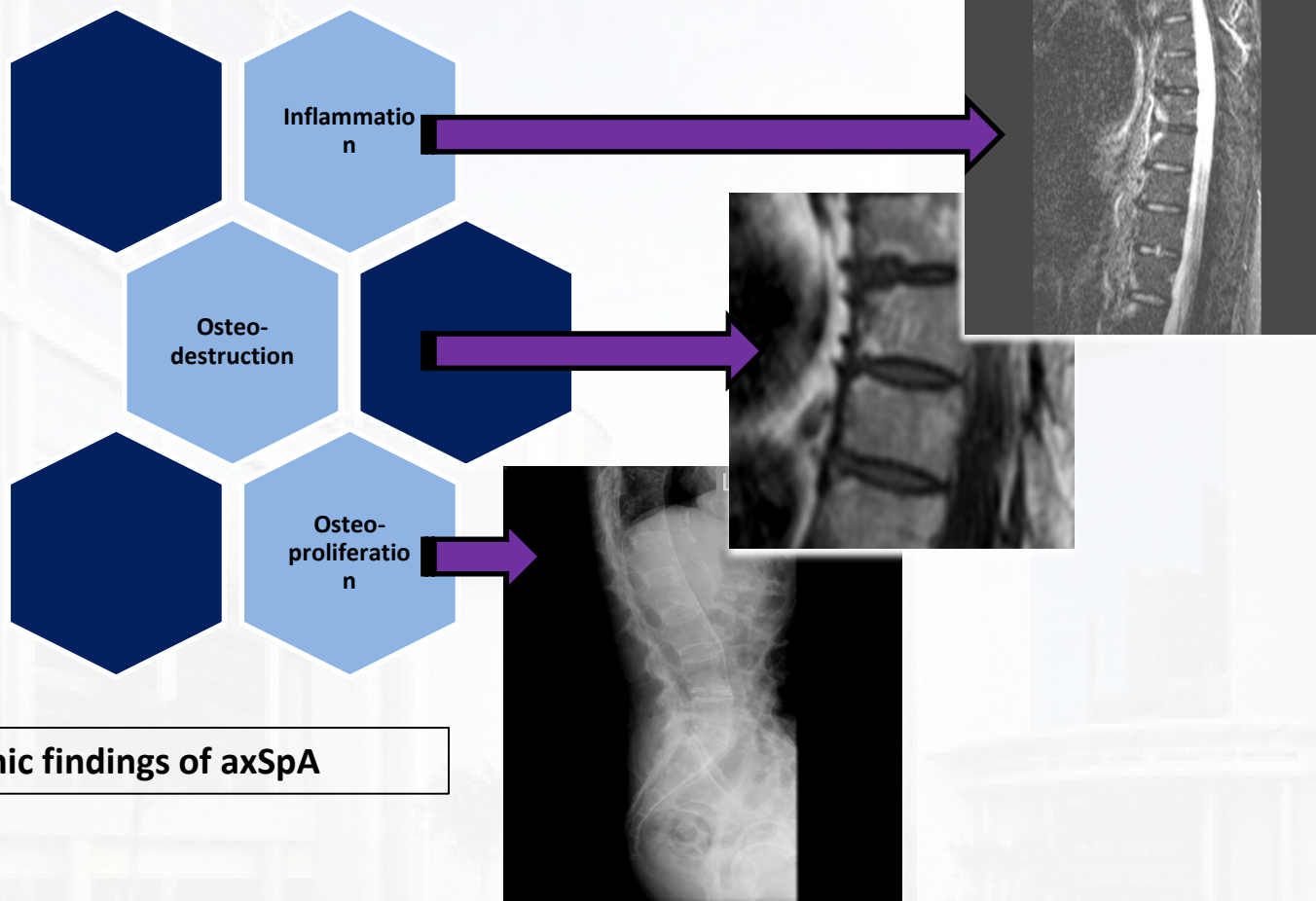
Three Different HLA-B27 Structures and Hypotheses as to How They Might Induce Disease Processes in AS



Final Stage of AS with Severe Kyphosis of Thoracic and Cervical Spine



Unable to look ahead while walking
(,patient cannot see the sun')



Pathognomonic findings of axSpA

Ankylosing Spondylitis: modified New York criteria

1. ***Clinical criteria***

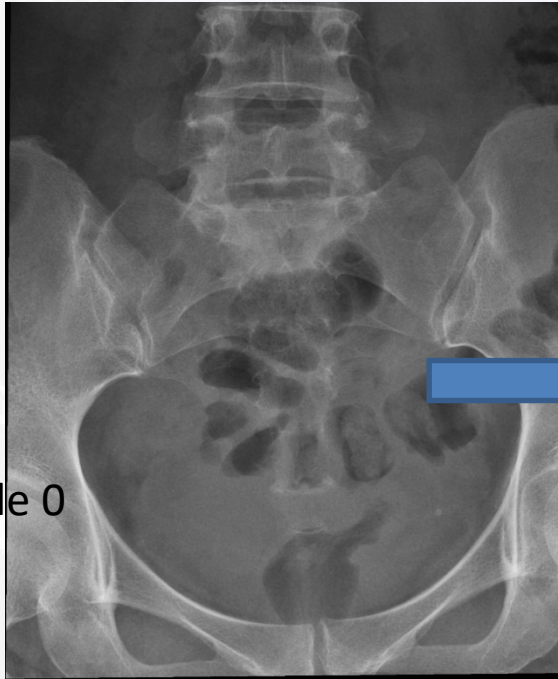
- a. Low back pain and stiffness for more than 3 months, which improves with exercise, but is not relieved by rest.
- b. Limitation of motion of the lumbar spine in both the sagittal and frontal planes.
- c. Limitation of chest expansion relative to normal values corrected for age and sex.

2. ***Radiological criteria***

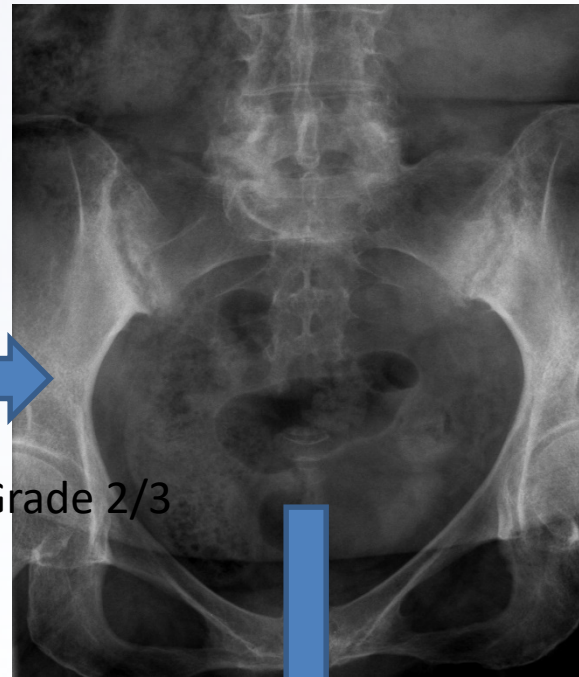
Sacroiliitis: grade >2 bilaterally or grade 3-4 unilaterally

Definite AS radiological criteria plus one clinical criteria

Grade 0

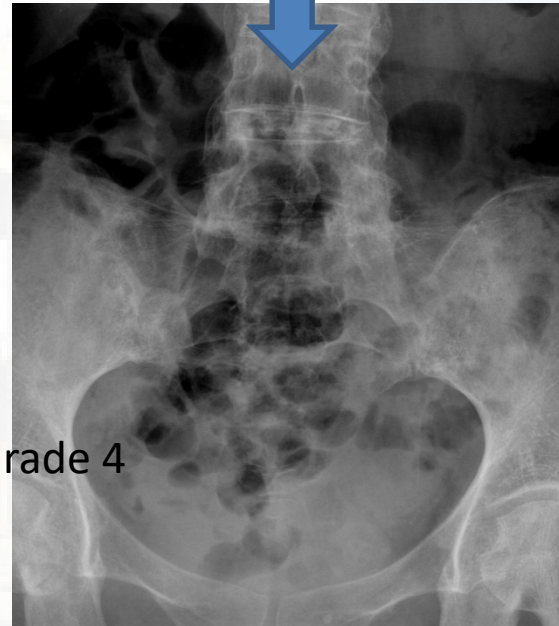


Grade 2/3



Radiological sacroiliitis

Grade 4



Use of imaging in axSpA

Disease Diagnosis

- MNY criteria
- ASAS classification criteria for axSpA



Diagnosis: X-ray sacroiliitis

Delay in diagnosis

- Up to 6-11 years^{1, 2}

¹Dincer, U., Cakar, E., Kiralp, M., & Dursun, Z. (2008). Diagnosis delay in patients with ankylosing spondylitis: Possible reasons and proposals for new diagnostic criteria. *Clinical Rheumatology*, 27(4), 457-462.

²Aggarwal, R., & Malaviya, A. (2009). Diagnosis delay in patients with ankylosing spondylitis: Factors and outcomes—an Indian perspective. *Clinical Rheumatology*, 28(3), 327-331.

ASAS Classification Criteria for Spondyloarthritis (SpA)

In patients with ≥ 3 months back pain and age at onset < 45 years

Sacroiliitis on imaging plus ≥ 1 SpA feature

OR

HLA-B27 plus ≥ 2 other SpA features

SpA features

- inflammatory back pain (IBP)
- arthritis
- enthesitis (heel)
- uveitis
- dactylitis
- psoriasis
- Crohn's/colitis
- good response to NSAIDs
- family history for SpA
- HLA-B27
- elevated CRP

In patients with peripheral symptoms ONLY

Arthritis or enthesitis or dactylitis plus

≥ 1 SpA feature

- uveitis
- psoriasis
- Crohn's/colitis
- preceding infection
- HLA-B27
- sacroiliitis on imaging

OR

≥ 2 other SpA features

- arthritis
- enthesitis
- dactylitis
- IBP ever
- family history for SpA

Sensitivity: 79.5%, Specificity: 83.3%; n=975

Disease activities



Bath Ankylosing Spondylitis Disease Activity Index

Patient no. _____

BASDAI

Bath Ankylosing Spondylitis Disease Activity Index
in Chinese language

BASDAI 活性量表：過去一週內，這些問題的嚴重程度，在每一條問題上標上

X 記號：

☐ 疲累的感覺

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

完全沒有

非常嚴重

☐ 脊椎（包含頸部、背部及髖關節）疼痛程度

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

完全沒有

非常嚴重

☐ 其他（頸部、背部及髖關節以外）關節疼痛程度

0	1	2	3	4	5	6	7	8	9	10
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完全沒有

非常嚴重

☐ 身體任何部位有觸痛或壓痛的不適程度

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

完全沒有

非常嚴重

☐ 早上起床晨間僵硬程度

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

完全沒有

非常嚴重

☐ 從起床開始計算，您的晨間僵硬時間

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

0小時

1小時

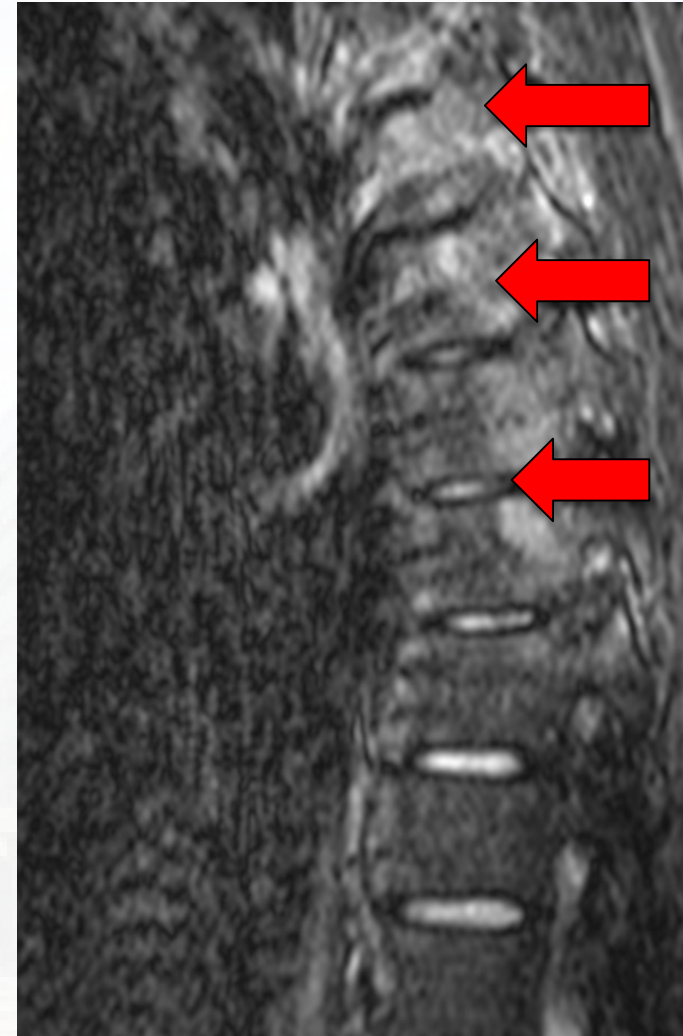
2小時

Bath Ankylos
Spondylitis Disease
Activity Index
(BASDAI)

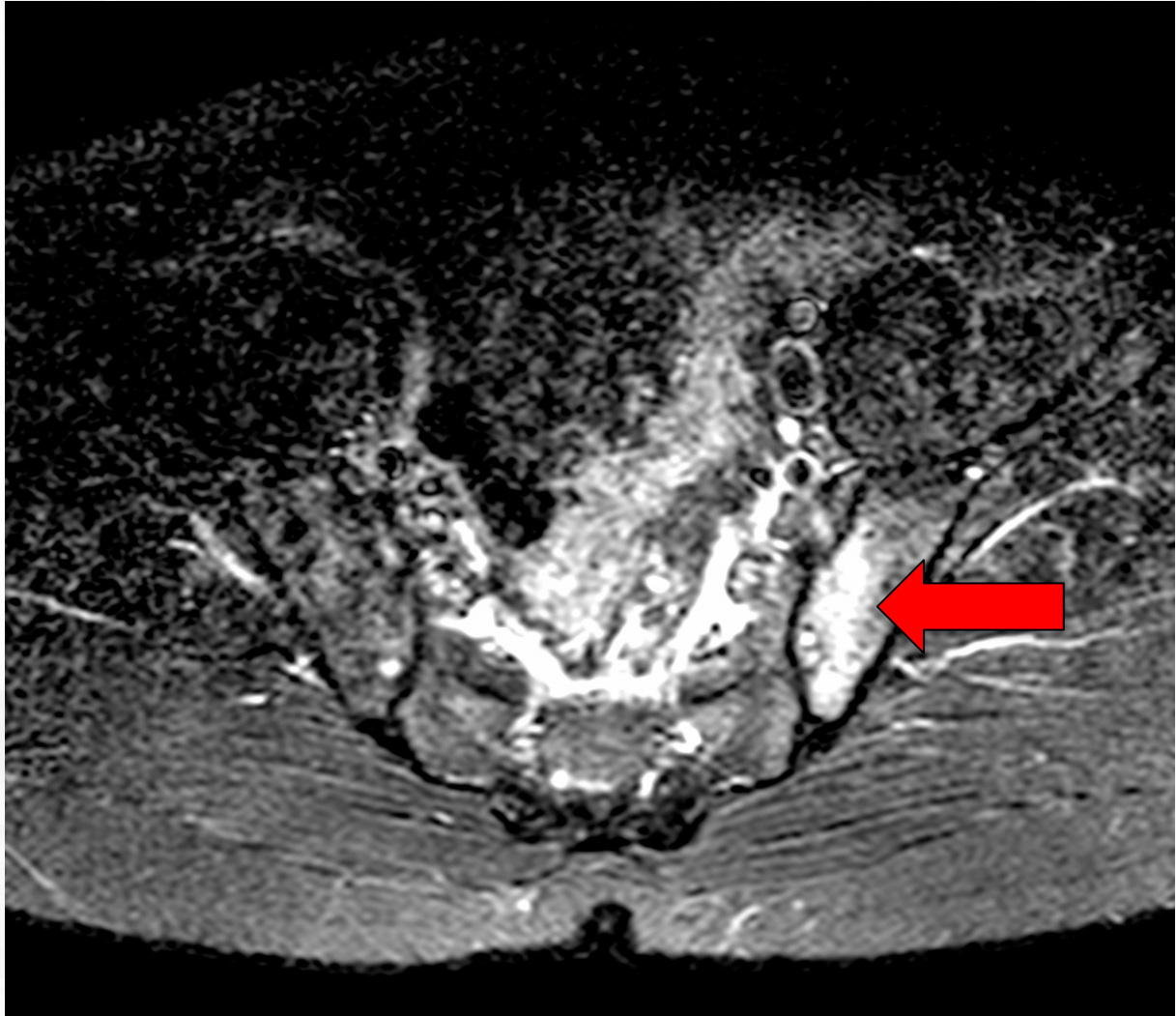
- Au YL, Wong WS, Mok MY, Chung HY, Chan E, Lau CS. Disease activity assessment in ankylosing spondylitis in a Chinese cohort: BASDAI or ASDAS? Clin Rheumatol. 2014 Aug; 33(8): 1127-34.

Using spine MRI to diagnose SpA

- ***Inflammation***
 - Anterior/ posterior spondylitis (corner inflammatory lesion [CIL])
 - Spondylodiscitis
 - Arthritis of costovertebral joints
 - Arthritis of zygoapophyseal joints
 - Enthesitis of spinal ligaments
- ***Structural changes***
 - Fatty deposition (fatty corner lesion [FCL])
 - Erosions
 - Syndesmophytes
 - Ankylosis



MRI SI joints --- STIR



Acute Arthritis of the Right Knee in a Patient with Peripheral Spondyloarthritis



Psoriasis



Typical psoriasis with scaly patches and plaques

Moderate to Severe Nail Changes in Patient with Psoriasis



Mild Nail Changes in Patient with Psoriasis



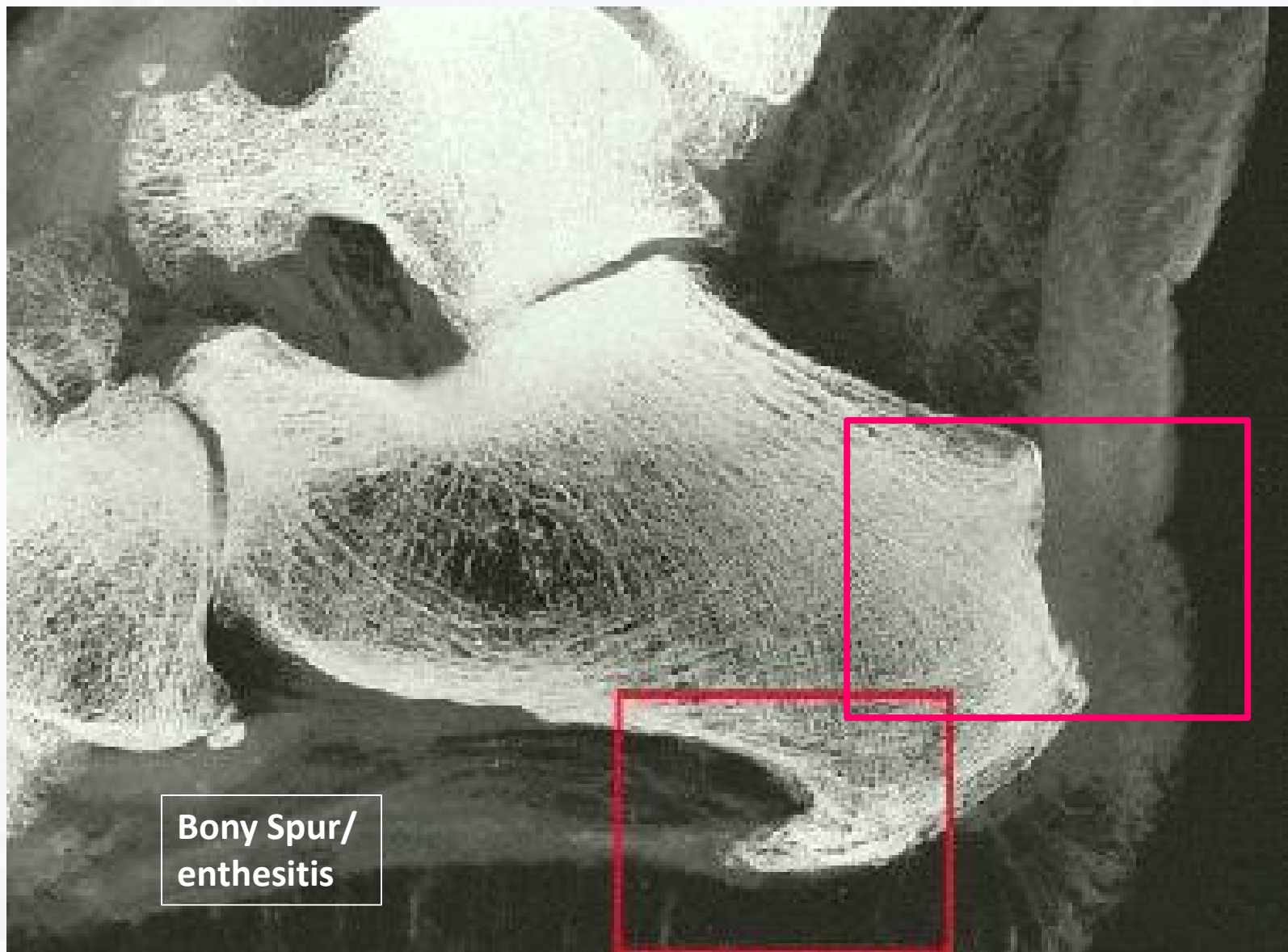
Dactylitis



* Psoriasis plaque lesions

Enthesitis (Insertion of Achilles Tendon at Calcaneus) Right Heel





Arthritis--- treatment goals

Goals:

- Suppress disease activity, prevent joint-damage and preserve normal function
- Pain relief



Arthritis --- treatment

- Education
- Physical therapy (Physio/ occupation therapy)
- Medical therapy
- Surgery

Arthritis --- medications

- Symptom-modifying anti-rheumatic drug
 1. Non-steroidal anti-inflammatory drug (NSAID) e.g. Diclofenac
- Disease-modifying anti-rheumatic drug (DMARD)
 1. Conventional synthetic DMARD (csDMARD) e.g. Methotrexate, sulfasalazine, leflunomide
 2. Biological DMARD (bDMARD) e.g. anti-TNF, anti-IL6, anti-IL17, anti-CD20
 3. Targeted synthetic DMARD (tsDMARD) e.g. Janus Kinase inhibitor --- tofacitinib
- Steroid

Orthopedic surgeries

Patient factors

- Location of the damaged joint
- Expectation of patient
- Age

Surgeon factors

- Availability of the expertise
- Previous experience of the surgeon







Acknowledgements:
➤ **Rheumatology team members**